

Research with AI as an undergraduate

What I could build.

What I could trust.

What I understand.

A neutrino analysis—and an experiment in how to do research.

Physics undergraduate · CS coterm · Entering fourth year

The starting point: an open analysis + OmniFold

“I nodded because I thought I understood.”

Public MINERvA analysis

Tutorial, data and a published 2D result

An unfolding method

Classifiers learn simulation weights

Several LLMs

Explain the pieces; help put them together

Simulation truth

→ detector →

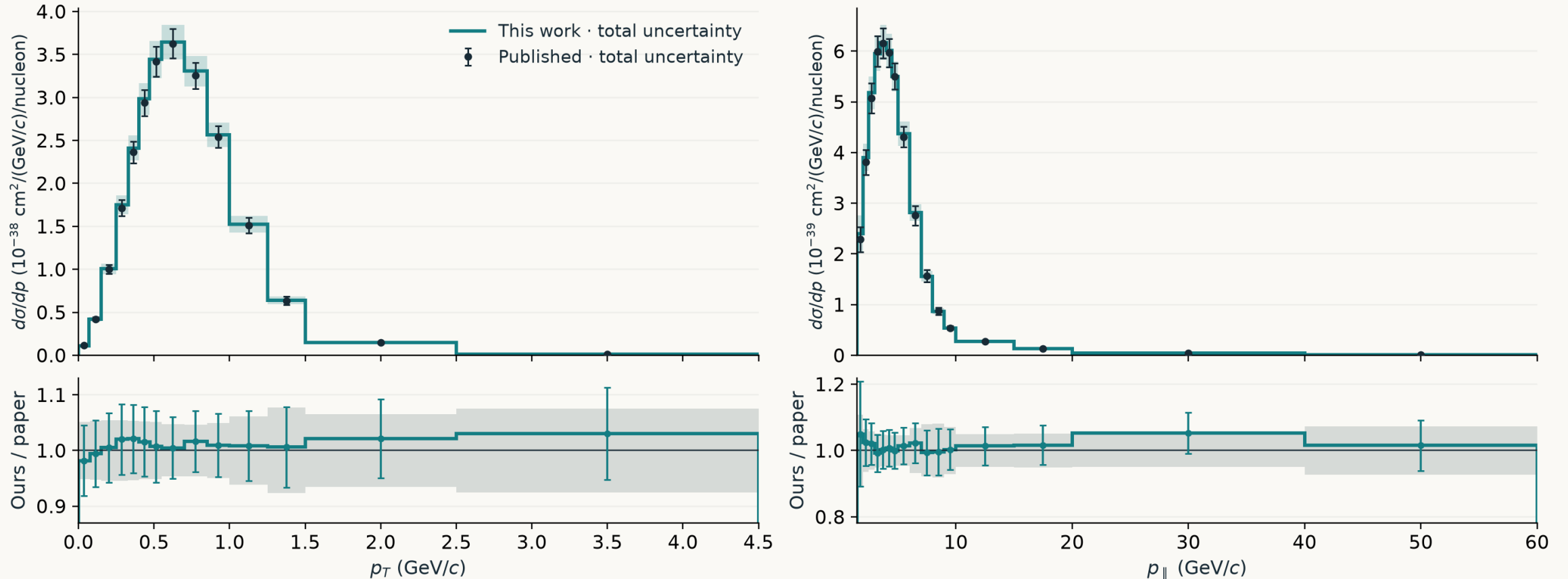
Reconstructed simulation

Transfer weights back

← classifier compares with data

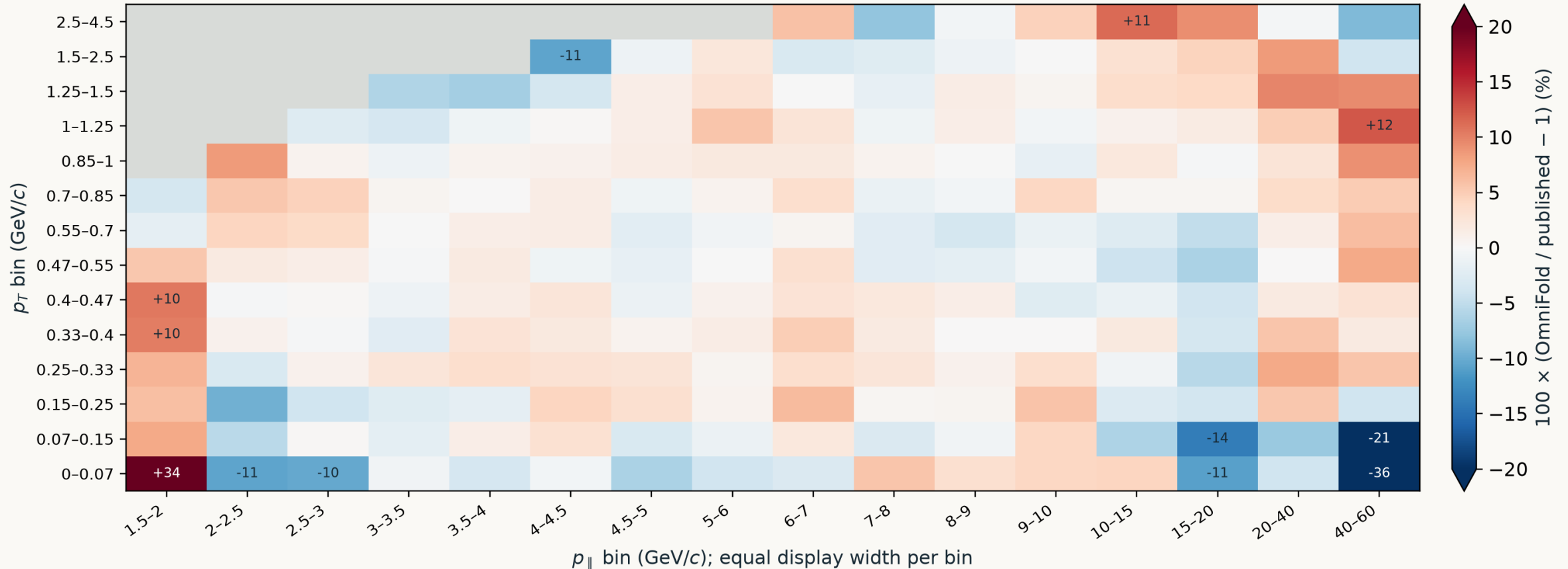
Initial target: inclusive ν_μ cross sections in transverse and longitudinal muon momentum.

How closely does the 2D result reproduce the paper?



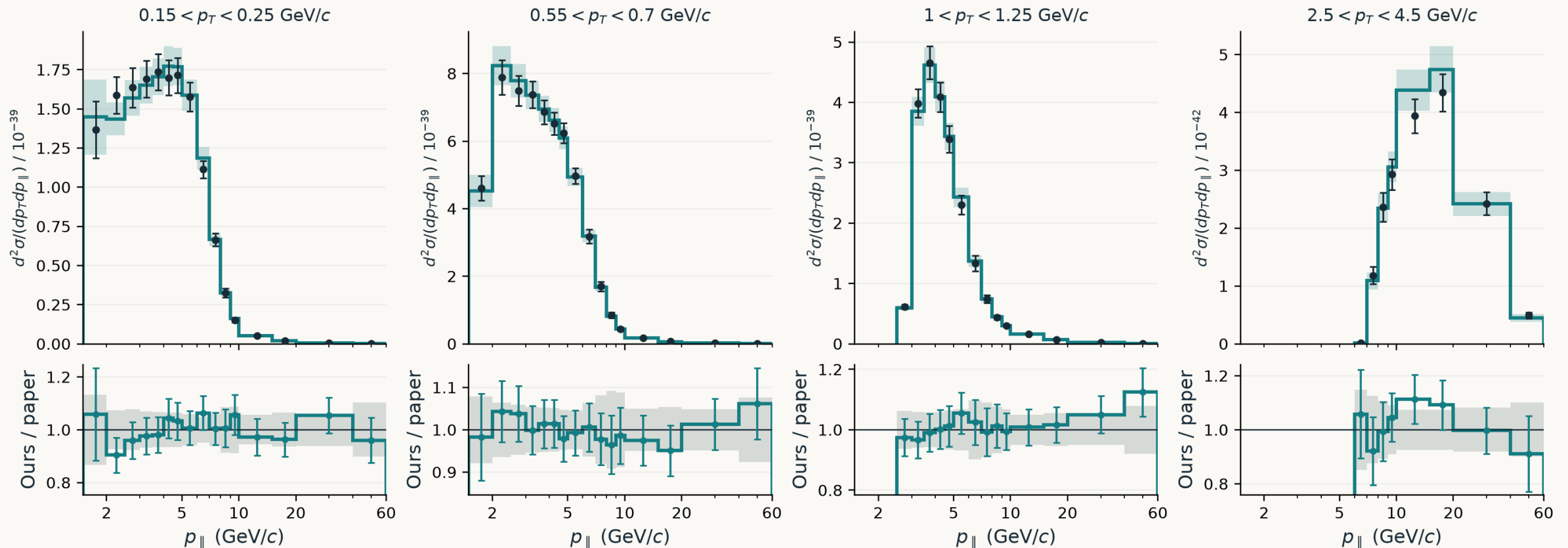
Teal: ours $\pm$ total error; black/gray: paper total error. Ratio bars: our error / paper central.
 Each uses its own bin widths; ratios match bins. Three p_T boundaries differ by 0.005 GeV/c.

The full 2D comparison shows where differences live



193 / 205 reported bins are within 10%; three differ by more than 20%. Gray = unreported.
Corresponding-bin comparison; y labels use this work's edges. Three p_T boundaries differ.

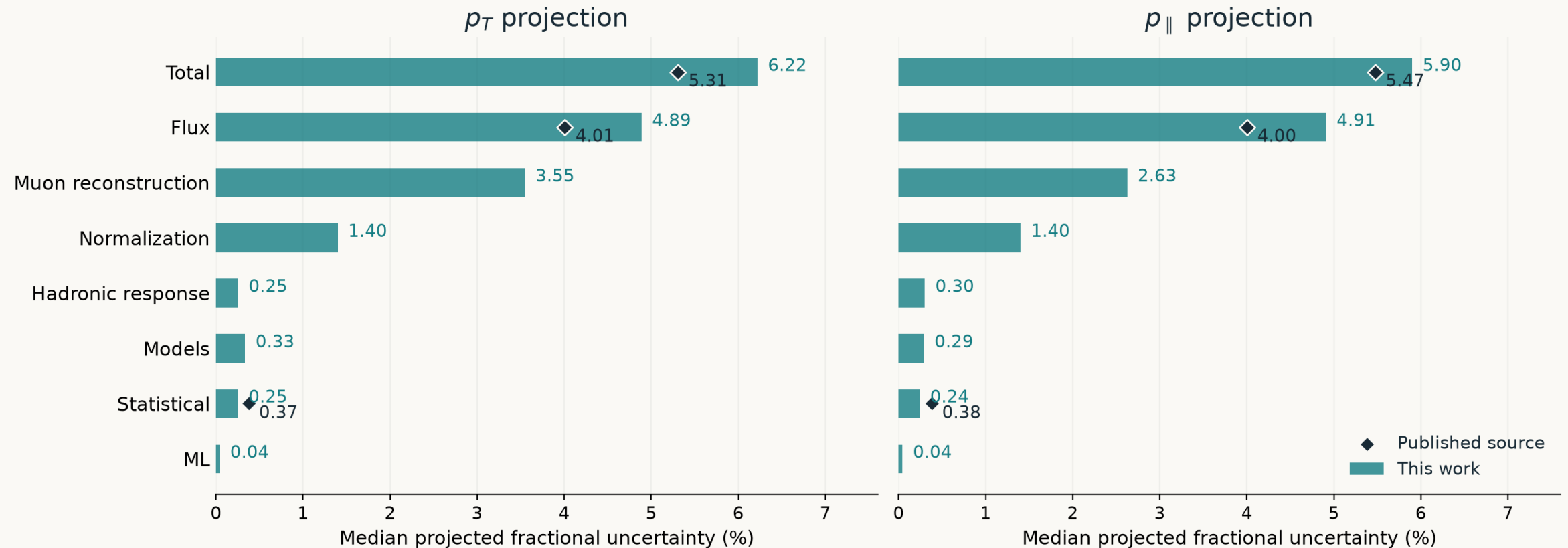
The agreement survives looking inside the projections



Four matched p_T slices; all 14 in backup. Density units: $\text{cm}^2/(\text{GeV}/c)^2/\text{nucleon}$.

Teal: ours $\pm$ total error; black/gray: paper total error. Ratio bars: our error / paper central.

Flux and muon reconstruction dominate our errors



Black markers show released paper components with matching scope; blanks do not mean zero.

Values are medians across projected bins, not additive shares. ML means unfolding-model seed variation.

Muon energy scale is a subset of reconstruction; its direct comparison is in backup.

The scope expanded beyond the published reference

2D

Muon momentum

**Central result +
uncertainty validated**

3D

+ available energy

**Central result +
closure validated**

4D / 5D

+ q_3 and W

**Central results +
closures validated**

Full event

Particle-cloud inputs

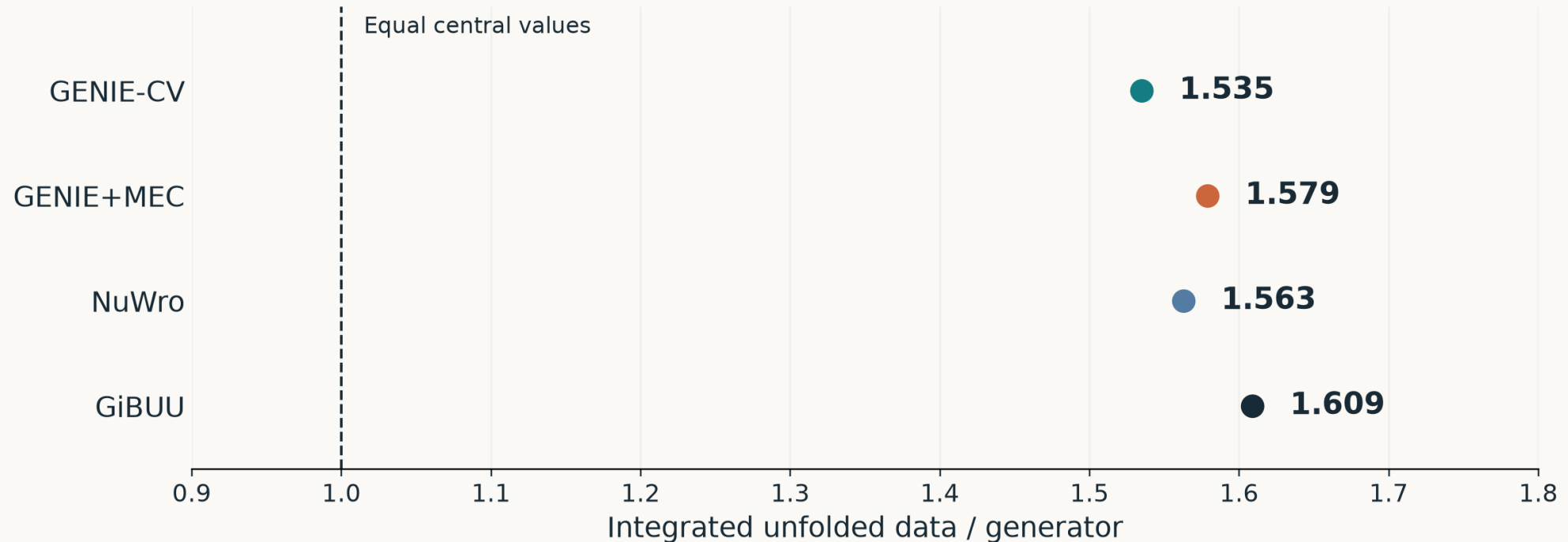
**PET diagnostics /
method development**

Higher-dimensional uncertainties remain unadopted.

Beyond reproduction, deciding what would count as validation becomes part of the research.

Beyond reproduction: a feature the generators miss

High available energy and high W: $E_{\text{avail}} \geq 0.8$ GeV, $W \geq 1.8$ GeV



Adding MEC does not close this gap: data / GENIE rises from 1.535 to 1.579.

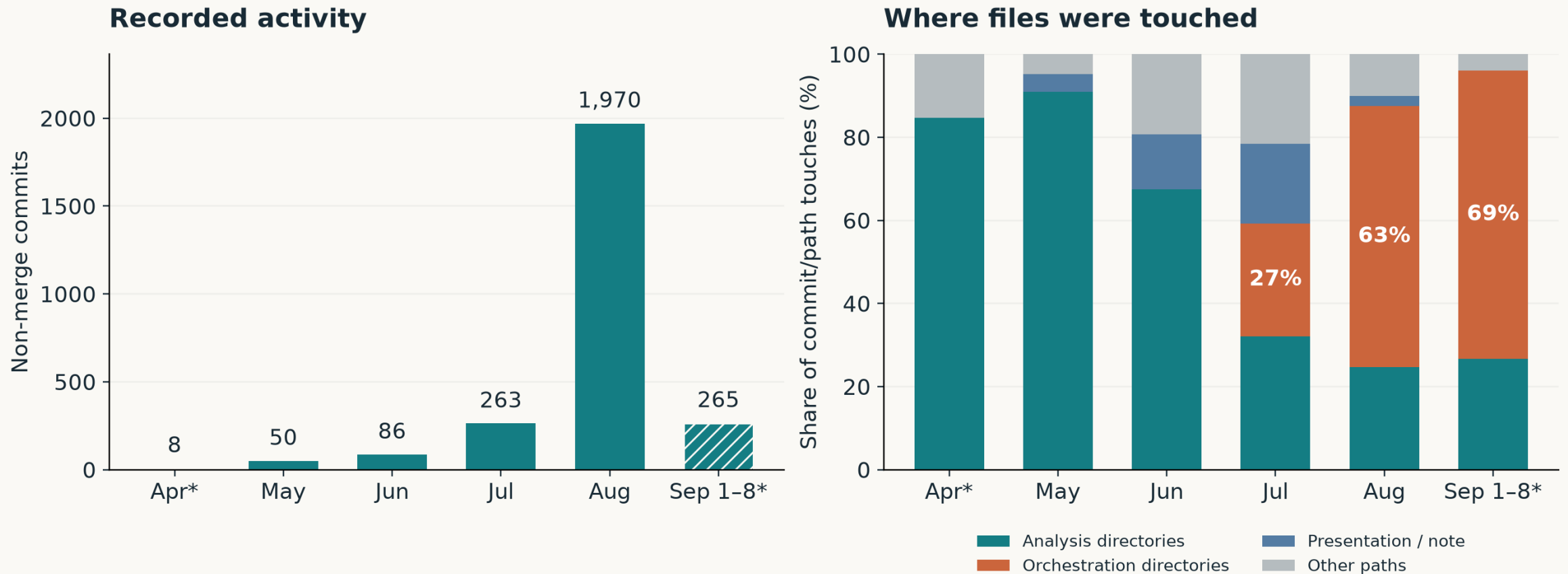
The significance awaits an adopted covariance; no uncertainty bars are shown.

The workflow became a project of its own

Start	Ask several models	Explain the tutorial; assemble a reproduction.
July 16–18	Coordinator + workers	Bounded tasks, separate verification and review.
July 19	Deterministic waking	Scripts detect events; models resume for work.
August 25	Campaign queue	Stage actions and checks; make execution explicit.

These are documented milestones, not clean experimental eras.

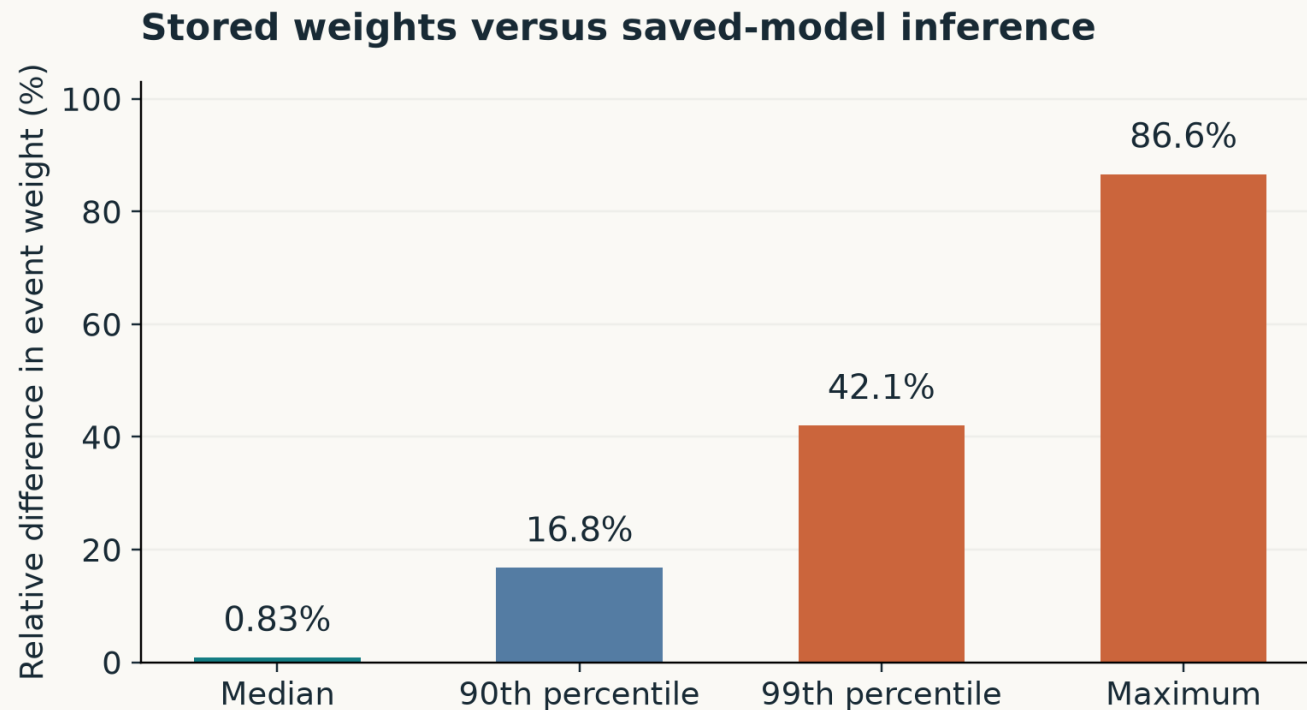
Activity grew. Was the work more effective?



The right panel measures edit locations. It does not measure hours or scientific value.

A saved model did not reproduce the event weights

Checkpoint = a saved snapshot of the network's learned parameters.



Used in the run

Last epoch
(final training pass)

Saved to disk

Best validation epoch
(lowest held-out loss)

Aggregate difference
< 0.0001

Learned model parameters → event weights → unfolded distribution

The friction: auditing before trying anything

What slowed me down

Agents wanted to audit everything before acting.

I often wanted a small trial-and-error experiment.

What I would change

For a cheap, reversible test: name the question and try it.

Before expensive runs or claims: check the evidence and provenance.

The question is when a check changes the next decision.

Proposed workflow change; no measured time saving is claimed.

My active time shifted into orchestration and routing

The work I remember doing

Debugging coordination,
account routing and wakeups.

The logs include wrong executable
paths and Python environments.

A simpler tool can help

“Why do we even need an
LLM for the watcher?”

A defined job outcome can be
reported by an ordinary script.

That overhead belongs in any account of productivity.

The record supports concrete failures; it does not supply a complete time budget.

Outside expertise corrected the question itself

The mistaken premise

A collaborator question said there was no prior MINERvA 3D unfolding result.

That premise was wrong.

The corrected framing

MINERvA already had a 3D unfolding publication.

The distinction here is the unbinned simultaneous method.

A working pipeline does not establish the novelty of its research claim.

This correction does not establish who introduced the error or how much AI contributed.

I have a high-level map. Do I understand the details?

My honest answer

AI abstracts so much away
that I am unsure what I could
explain without it.

I can see the overall approach.

The tradeoff I want to test

Abstraction might help me avoid
getting stuck in a rabbit hole.

It might also hide a mistake
I cannot recognize or explain.

A useful check: predict a change before asking the model.

Then explain the result and trace the evidence without relying on the model's summary.

Real research output; an unresolved net benefit

What I would keep

Freely asking basic questions.
Help implementing unfamiliar work.
External references and closure
as tests of research outputs.

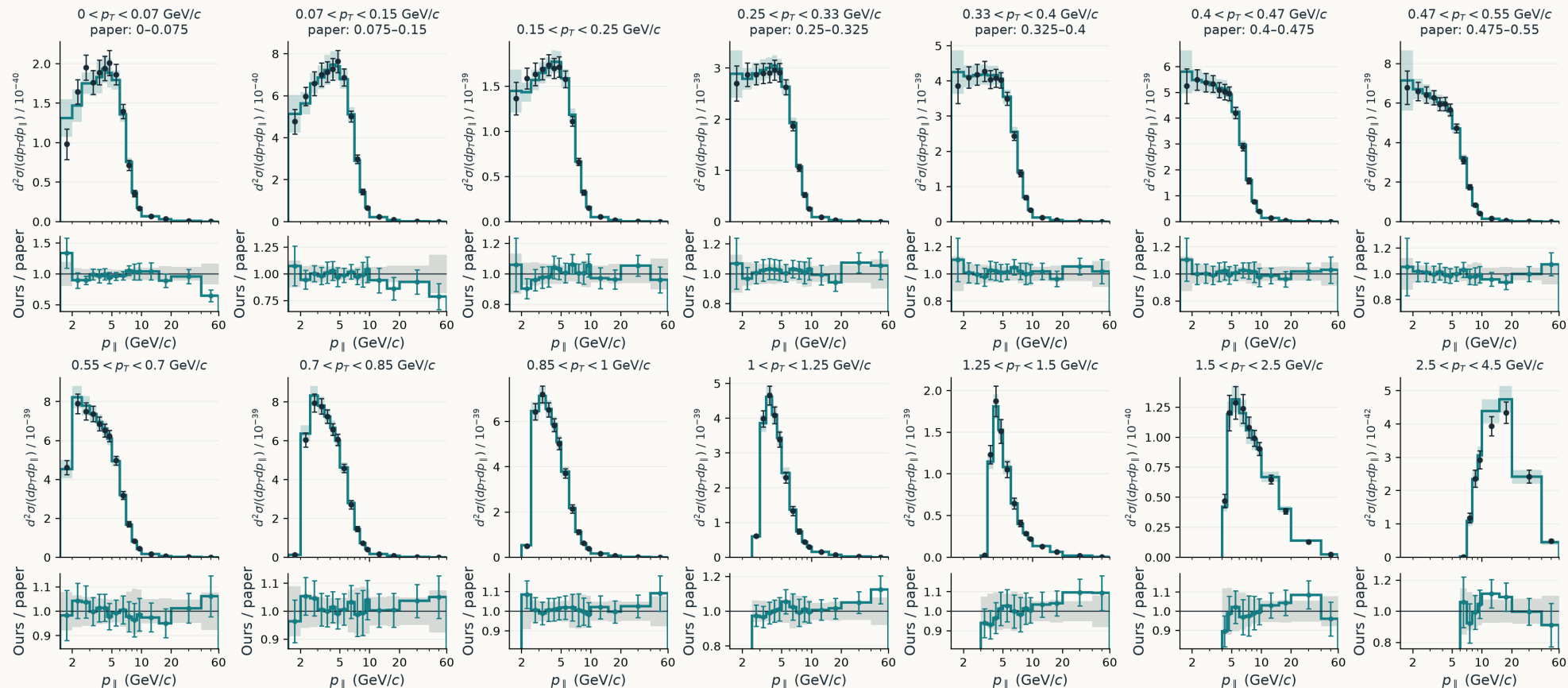
What I would change

Smaller, bounded experiments.
Simpler recurring operations.
Track my intervention time.
Practice explaining key decisions.

What should a student demonstrate before delegating this much?

Judge the accepted research result, the total effort, and the student's understanding separately.

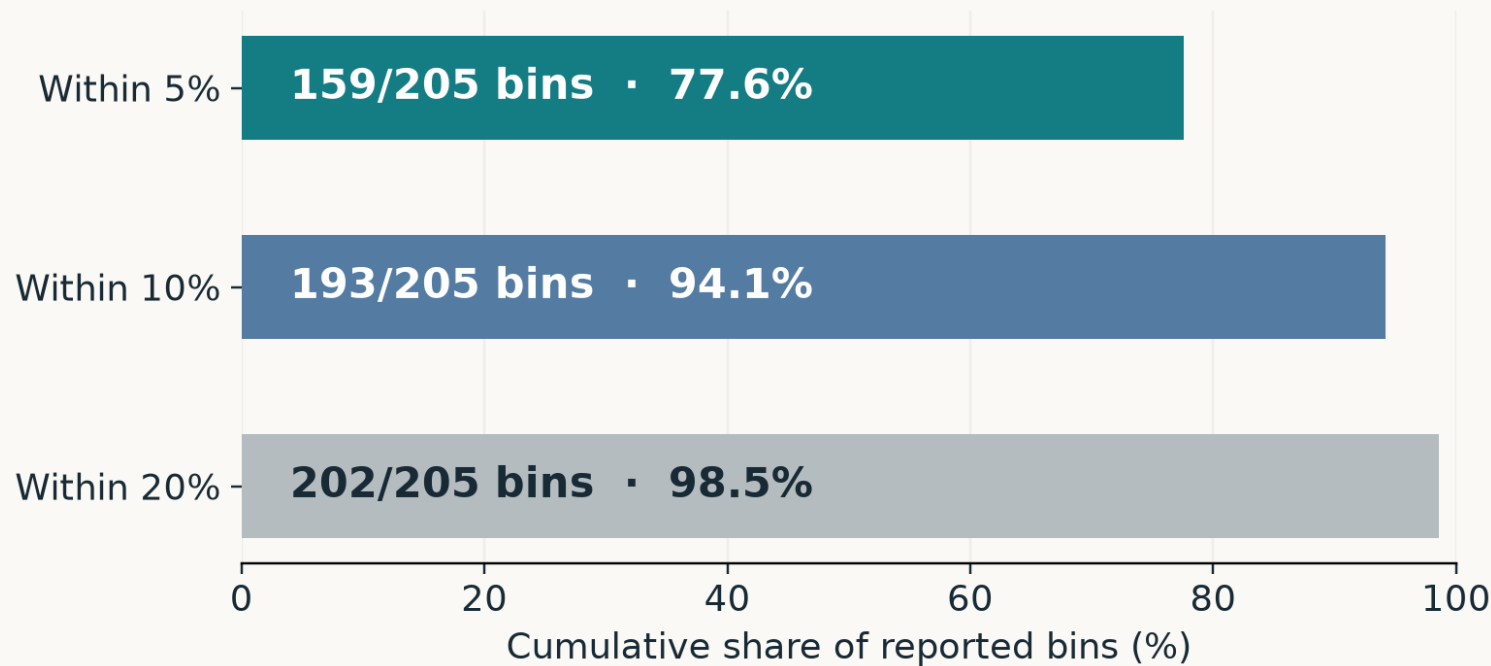
All 14 transverse-momentum slices



Full slice inventory; differing p_T boundaries labeled. Teal: our total error; black/gray: paper total error.
Density units: cm²/(GeV/c)²/nucleon; per-panel powers of ten are printed on the axes.

The 2D reproduction supplied an external reference

How close are the central values?



Integrated cross section

3.073 this work
3.039 published

$\times 10^{-38} \text{ cm}^2 / \text{nucleon}$

About 1.1% apart

Completed checks: closure, completeness, iterations and standalone uncertainty.

A reassuring ratio can hide the wrong quantity

A differential cross section needs bin areas to become a total.

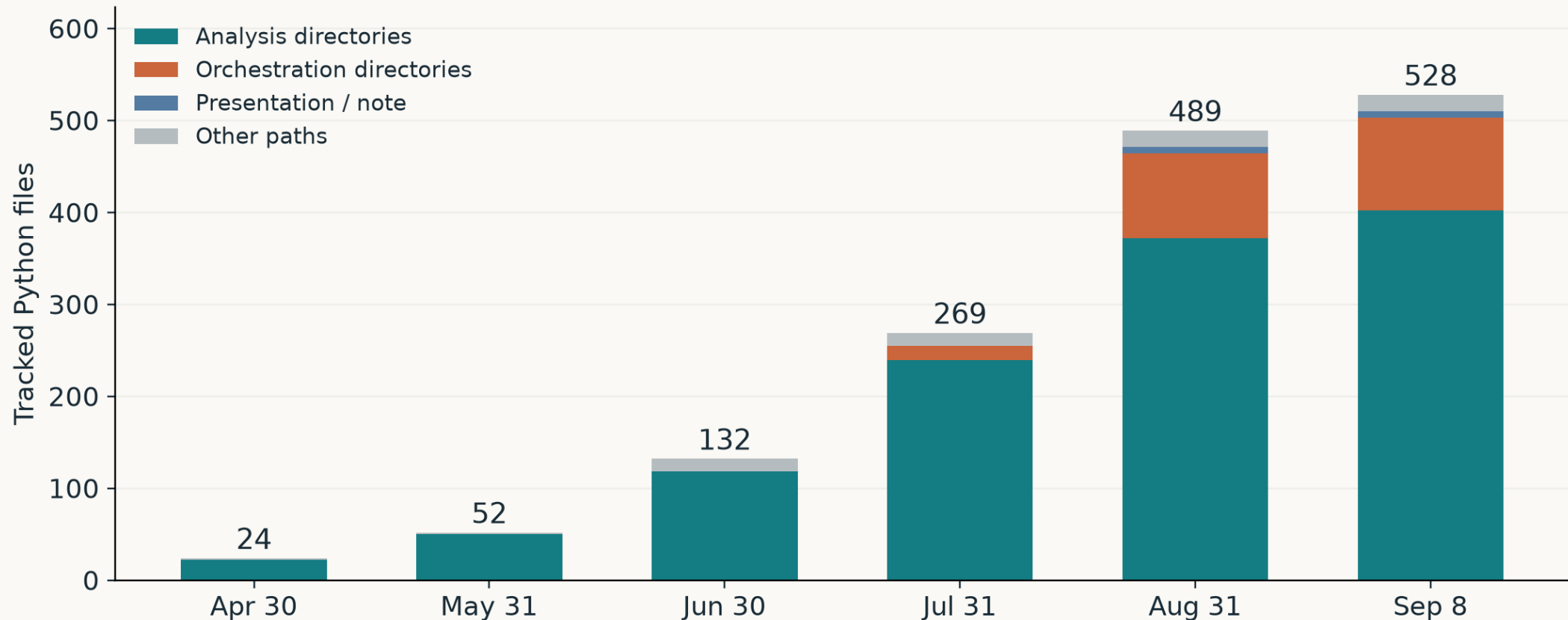
$$\sigma = \sum_{i,j} \left[\frac{d^2\sigma}{dp_T dp_{\parallel}} \right]_{ij} \Delta p_{T,i} \Delta p_{\parallel,j}$$

Calculation	This work	Ours / paper
Integrated total	3.073×10^{-38}	1.0113
Bare sum of densities	3.732×10^{-37}	1.0115

Only the integrated row has total-cross-section units: cm²/nucleon.

Check the definition and units—not only agreement with another number.

Repository size describes scope, not understanding



File-edit entropy describes concentration of edits. It is not a code-complexity score.

What this case study can establish

Available evidence

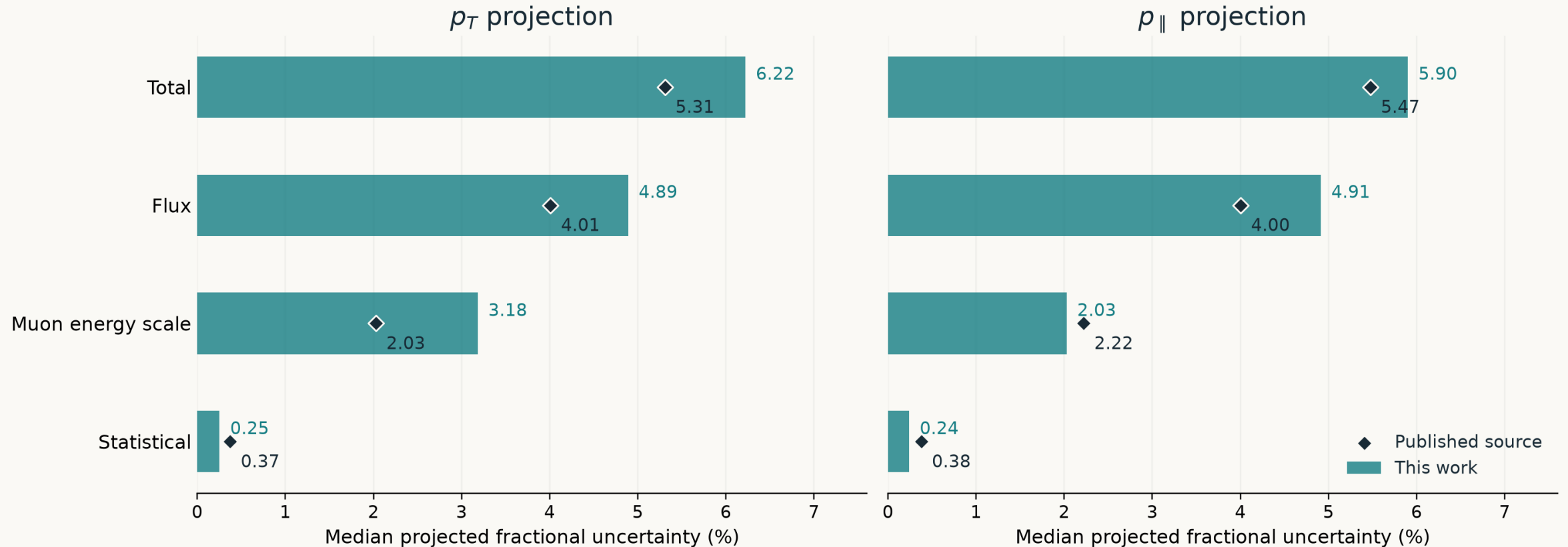
- Completed research components
- Historical failure / repair pairs
- Repository activity and scope
- The speaker's own recollections

Not measured

- A matched no-AI baseline
- Human hours saved
- A controlled model ranking
- A learning score

Release dates and commit counts alone cannot identify model-specific productivity gains.

Compare the sources released with the paper



Our energy-scale group is MINOS + MINERvA; it excludes efficiency, resolution and beam-angle terms.
 Each input uses its own bin widths. Similar source sizes do not establish identical covariance constructions.

Research references and scope

MINERvA inclusive CC measurement

Ruterbories et al. · [arXiv:2106.16210](#)

OmniFold

Andreassen et al. · [arXiv:1911.09107](#)

Unbinned-unfolding software

Milton et al. · [github.com/rymilton/unbinned_unfolding](#)

OmniLearn / PET

Mikuni & Nachman · [arXiv:2404.16091](#)

MINERvA foundation-model work

Krzmanec et al. · [arXiv:2604.12364](#)

2D validated · Higher-D covariance unadopted · PET diagnostic / method development